DP Biology – B4.2

Ecological niches

- Why does it help organisms to have specialized ways of feeding?
- How do a species' adaptations relate to its role in an ecosystem?

SL statements B4.2.1–13. Textbook pp. 363–379.
Image: a mossy leaf-tailed gecko, camouflaged. Charles J. Sharp, CC BY-SA 4.0.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	B4.2.1–6	Define a niche, compare oxygen needs, and compare photosynthetic, holozoic, mixotrophic and saprotrophic nutrition
2	B4.2.7–10	Describe archaea, infer hominid diets from teeth, and explain adaptations of herbivores, plants, predators and prey
3	B4.2.11–13	Explain how forest plants reach light, distinguish fundamental and realized niches, and explain competitive exclusion

The big picture: every species has a job

Feeding	Tolerance	Interactions
How it gets carbon compounds	Oxygen, light, temperature	Predators, prey, competitors

A **niche** is a species' role. Specialized ways of feeding let species **share** an ecosystem rather than compete head-on.

Big question
Why did grey squirrels push red squirrels out of most of Britain?

What is a niche?

B4.2.1

Habitat: where it lives

An address

Niche: how it lives

A job: food, enemies, activity times, tolerances

Discuss

Describe the niche of an organism you know: what it eats, what eats it, and one abiotic limit.

Oxygen and three groups of organisms

B4.2.2

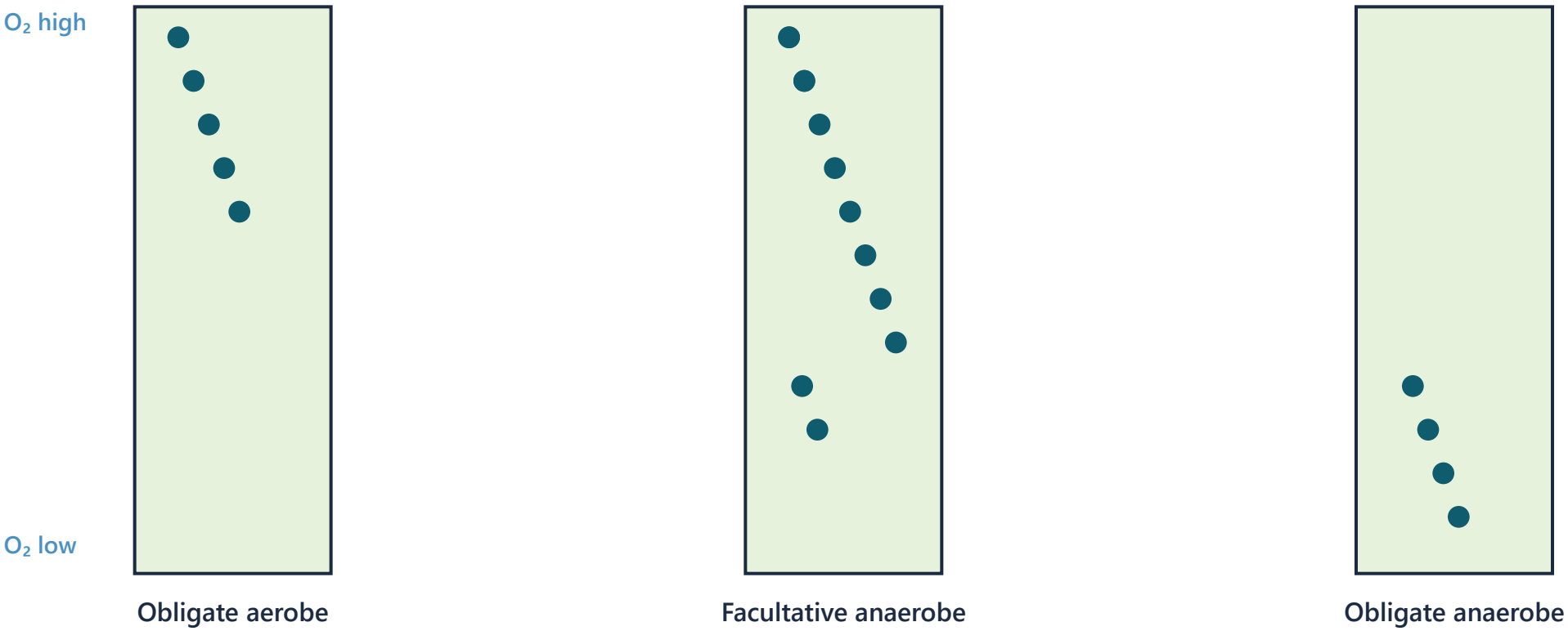
Group	Oxygen	Example
Obligate aerobe	Needs it	Humans, most plants
Facultative anaerobe	With or without	Yeast, E. coli
Obligate anaerobe	Cannot tolerate it	Clostridium tetani

Key idea

Oxygen tolerance is part of a niche. Tetanus bacteria live where there is no oxygen, such as deep puncture wounds.

Where do they grow in a tube?

B4.2.2



Read it

Aerobes grow at the top; anaerobes at the bottom; facultative anaerobes throughout, most at the top.

Modes of nutrition

B4.2.3–6

Mode	How carbon compounds are obtained	Examples
Photosynthetic	Made using light, from CO_2 and water	Plants, algae, cyanobacteria
Holozoic	Ingested, digested inside, absorbed	Most animals
Mixotrophic	Both photosynthesis and feeding	Euglena
Saprotrophic	Enzymes onto dead matter; absorbed	Many fungi, bacteria

Common mistake

Fungi are not plants and do not photosynthesize.

Holozoic nutrition

B4.2.4



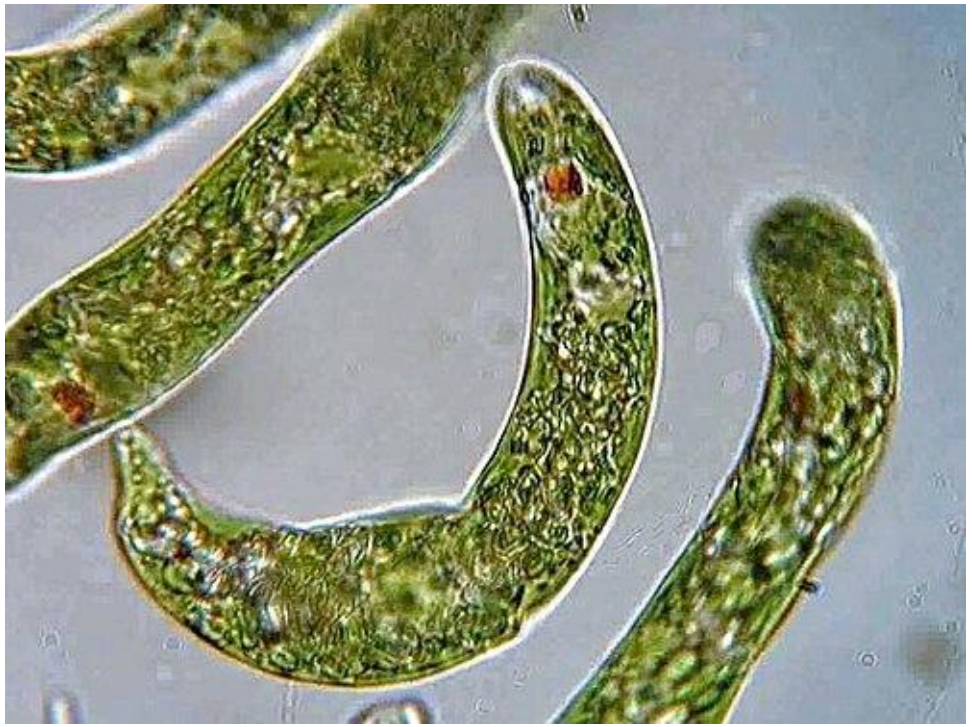
Food is taken in, broken down **inside** a gut, absorbed into blood and built into tissues; undigested waste leaves.

Amoeba does the same inside food vacuoles.

Photosynthesis

Plants, algae (such as kelp) and several prokaryote groups make their own food from light.

Mixotrophic nutrition



Euglena. Deuterostome, CC BY-SA 3.0.

Euglena photosynthesizes in light and feeds on particles too.

Type	Meaning
Facultative	Can use either or both modes
Obligate	Must use both

Discuss

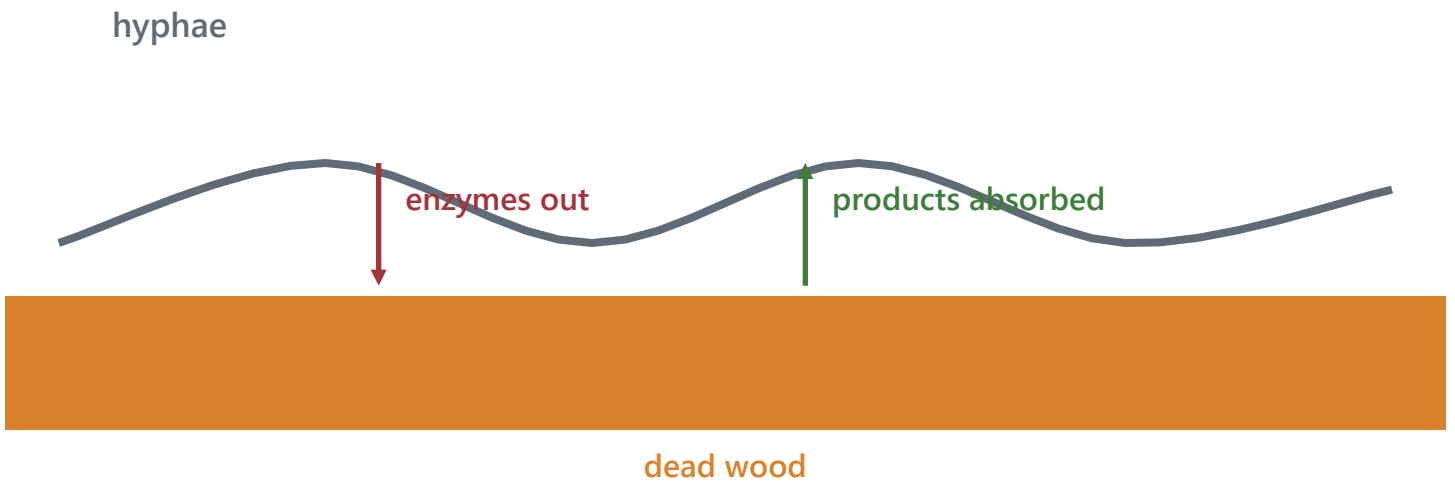
Advantage in a dim pond full of bacteria? What does a mixotroph lose compared with a specialist?

Saprotrophic nutrition

B4.2.6



Fungi on wood. Bernard Spragg, public domain.



Key idea

Digestion is **outside** the cells. Decomposers recycle elements (C4.2).

Practice: how do they feed?

B4.2.1–6

Classify

Mushroom, kelp, Euglena, lion

Explain

Why tetanus is a risk in deep wounds.

Distinguish

Habitat and niche.

Use

holozoic, saprotrophic, mixotrophic, obligate anaerobe

Check-in: class 1

B4.2.1–6

1. Define niche.
2. Name three groups by oxygen need.
3. List the stages of holozoic nutrition.
4. Define mixotroph, with an example.
5. Describe saprotrophic nutrition.

How to use this

Answer without your notes, then check against slides 3–11.

Diversity of nutrition in archaea

B4.2.7



Grand Prismatic Spring. Jim Peaco, NPS, public domain.

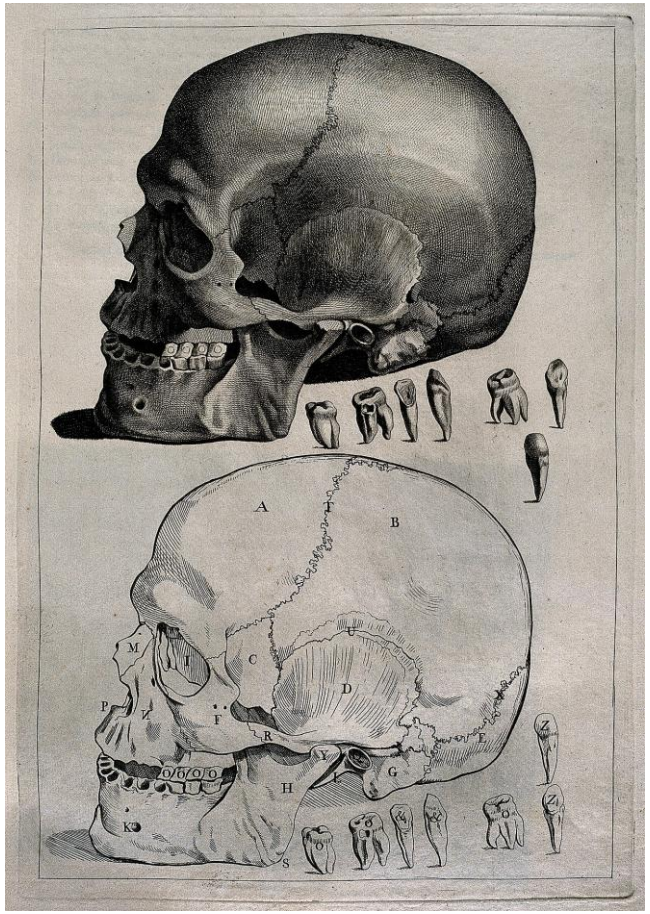
Nutrition	How
Photosynthetic	Light, often with non-chlorophyll pigments
Chemosynthetic	Energy from oxidizing inorganic chemicals
Heterotrophic	Carbon compounds from others

Common mistake

Archaea are not bacteria: they form a separate domain.

Teeth and diet in the Hominidae

B4.2.8



Human skull. Wellcome, CC BY 4.0.

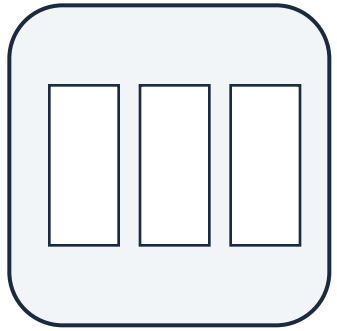
Species	Teeth	Likely diet
Homo sapiens	Small teeth	Omnivore
Homo floresiensis	Large premolars, strong jaw	Tough plants, some meat
Paranthropus robustus	Very large flat molars	Tough plants

Practice

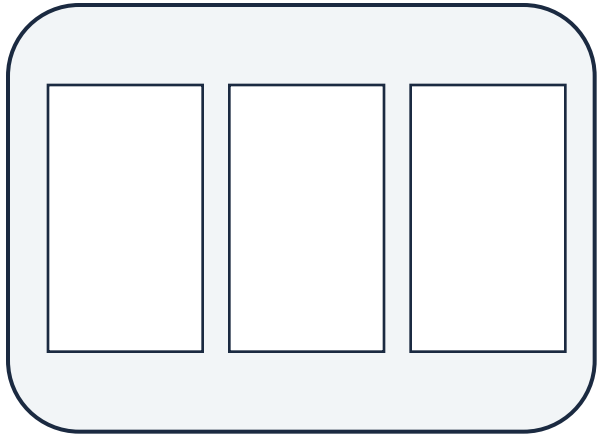
Use 3D skull models to infer diet. What do digital models miss?

Reading teeth

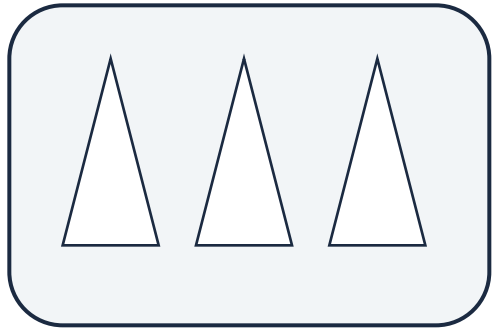
B4.2.8



small molars: mixed diet



huge flat molars: grinding plants



pointed teeth: meat

Key idea

Tooth size, shape and jaw strength give evidence of diet, even from fossils.

Herbivores: adapted to eat plants

B4.2.9



Monarch caterpillar. Public domain.

Chewing insects eat leaves; **piercing** insects (aphids) tap phloem.

Ruminants have flat grinding teeth and gut microbes to digest cellulose.

Aphids

A thin stylet secretes pectinase to slip between cells to the phloem; pressurized sap flows in.

Plants: resisting herbivores

B4.2.9



Foxglove. Public domain.



Castor bean plant. Public domain.

Physical: spines, stinging hairs.

Chemical: toxins; foxglove affects the heart; castor seeds contain ricin.

Arms race

Some herbivores evolve ways to detoxify; plants and herbivores evolve in response to each other.

Practice: archaea, teeth and herbivores

B4.2.7–9

Explain

How archaea at deep vents get energy.

Infer

Diet from huge flat molars and a crest on the skull.

Explain

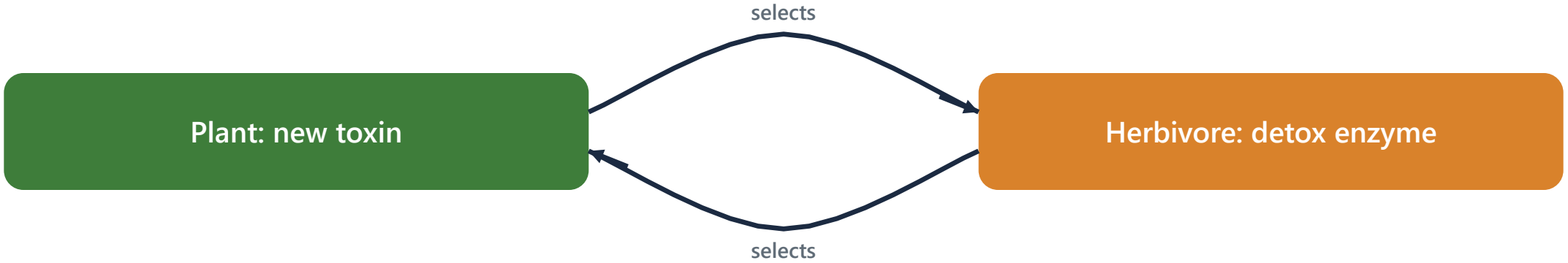
Why some insects can eat toxic plants.

Use

chemosynthesis, dentition, secondary compounds, coevolution

An evolutionary arms race

B4.2.9–10



Same pattern

Predators and prey: faster cheetahs, faster gazelles; better camouflage, sharper eyes. Each side's adaptation is the other side's selection pressure.

Predators and prey

B4.2.10



Wolf. NPS, public domain.



Leaf-tailed gecko. C. J. Sharp, CC BY-SA 4.0.

Type	Predators	Prey
Physical	Teeth, claws, keen senses	Camouflage, shells, spines
Chemical	Venom	Toxins, warning colours
Behavioural	Group hunting, ambush	Groups, alarm calls, hiding

Activity

For a local predator and prey, suggest one adaptation of each type.

Working with data: predator and prey numbers

B4.2.10

Hares and lynx trapped (thousands) in a northern forest. Rounded, based on the classic Hudson's Bay pattern.

Year	Hares	Lynx
1	20	2
3	80	4
5	40	8
7	10	3
9	30	2

1. Describe the pattern.
2. Explain why lynx peaks follow hare peaks.
3. Suggest another factor affecting hares.

Check-in: class 2

B4.2.7–10

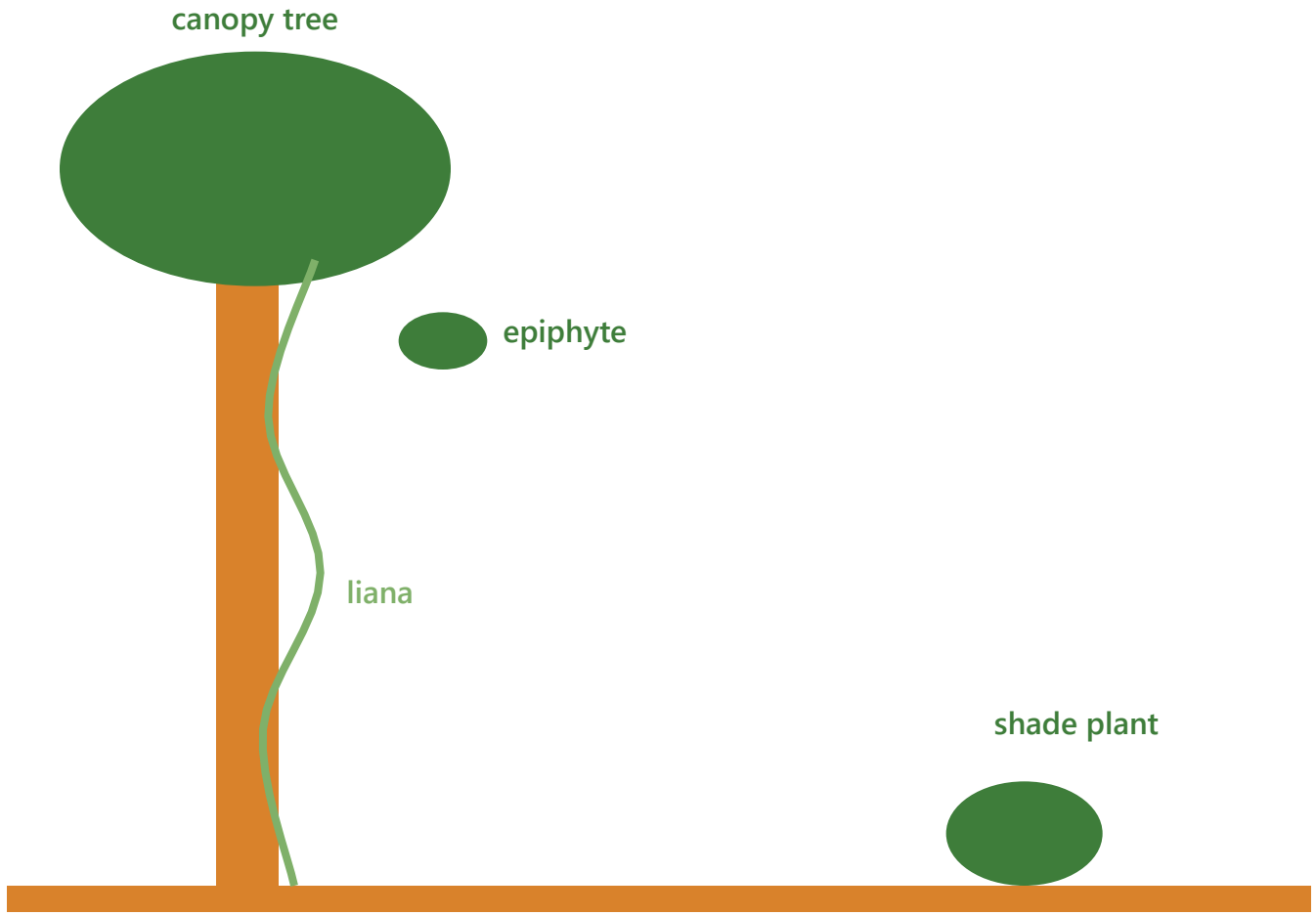
1. Give three modes of nutrition in archaea.
2. Explain how teeth show diet.
3. Give two herbivore adaptations.
4. Give a physical and a chemical plant defence.
5. Give one predator and one prey adaptation of each type.

How to use this

Answer without your notes, then check against slides 13–21.

Reaching light in a forest

B4.2.11



Form	Strategy
Canopy tree	Tall trunk
Liana	Climbs trees
Epiphyte	Grows on branches
Strangler fig	Starts high, roots down
Shade plant	Copes with low light

Climbers and perchers

B4.2.11



Monkey ladder vine. Gh5046, public domain.



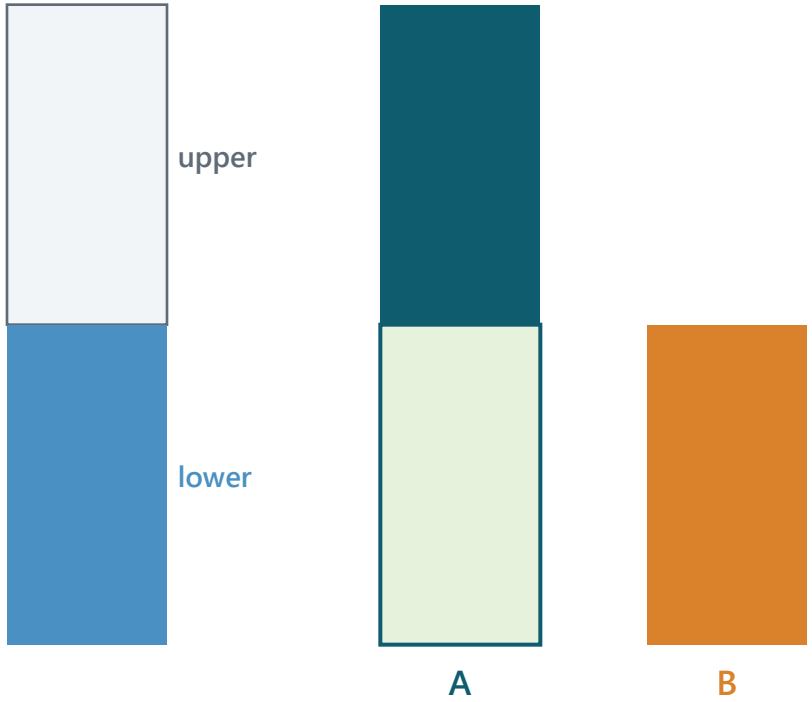
Epiphytes on a tree. James St. John, CC BY 2.0.

Key idea

Competition for light produces forest layers, each with its own niches.

Fundamental and realized niches

Niche	Meaning
Fundamental	Potential range, from adaptations and tolerance limits
Realized	Actual range when competing with others



Barnacles

A could live high and low (outline = fundamental), but B outcompetes it low down, so A is found only high (solid = realized).

Working with data: two barnacles

B4.2.12–13

Where two barnacles live, with and without each other (rounded, after Connell's study).

Situation	Upper shore	Lower shore
A alone	Present	Present
B alone	Absent	Present
Both	A only	B only

1. Describe each fundamental niche.
2. Explain A's realized niche.
3. Name the principle.

Competitive exclusion

B4.2.13



Red squirrel. Jojo, CC BY-SA 3.0.



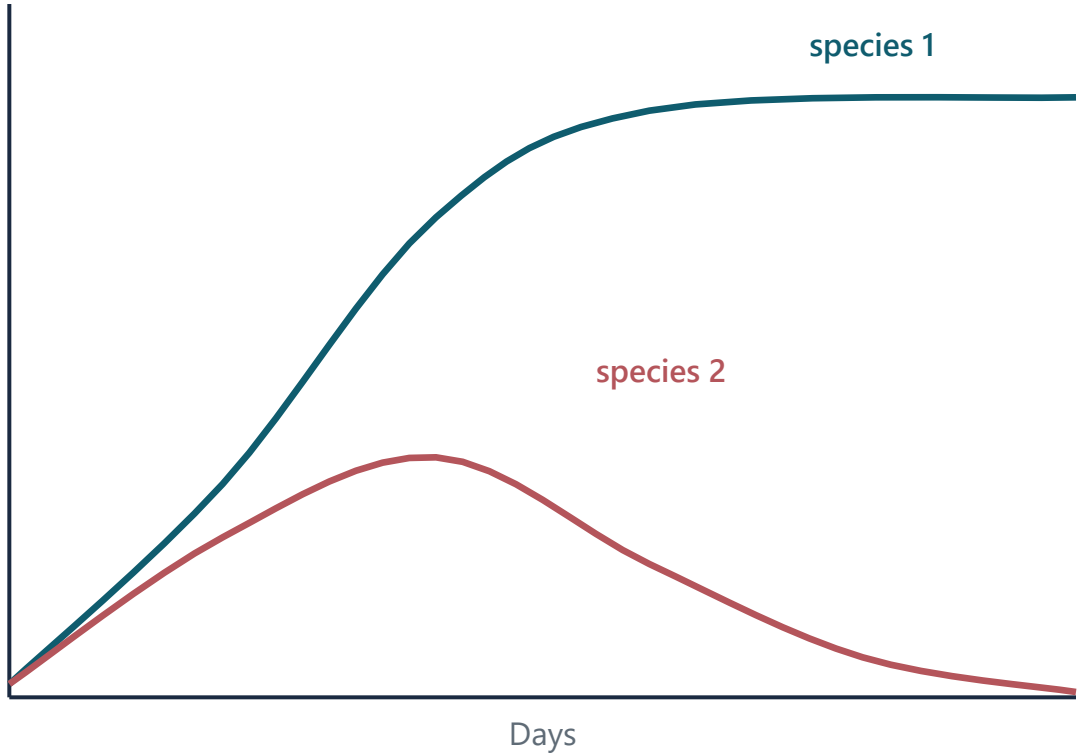
Grey squirrel. PierreSelim, CC BY 3.0.

Two species cannot share one niche for long: one is **eliminated**, or both are **restricted**.

Big question

Grey squirrels from North America replaced most red squirrels in Britain, except in some northern conifer forests.

Paramecium in a flask



Grown together. Sketch of the general pattern.

Read it

Alone, both thrive. Together, species 1 outcompetes species 2, which dies out.

Practice: niches and competition

B4.2.11–13

Explain

How an epiphyte reaches light.

Distinguish

Fundamental and realized niche.

Predict

Remove grey squirrels from a wood.

Use

canopy, realized niche, competitive exclusion

Check-in: class 3

B4.2.11–13

1. Give three ways forest plants reach light.
2. Distinguish fundamental and realized niches.
3. State the principle of competitive exclusion.
4. Give two possible outcomes of competition.
5. Outline the red and grey squirrel example.

How to use this

Answer without your notes, then check against slides 23–29.

Common mistakes to avoid

Not this	But this
A niche is a habitat.	Habitat is where; niche is the whole role.
Fungi are plants.	Fungi are heterotrophs; many are saprotrophs.
Archaea are bacteria.	Archaea are a separate domain.
Two species can share one niche forever.	One will exclude the other.

Summary: ecological niches

	Key idea
B4.2.1–2	Niche; oxygen needs
B4.2.3–6	Photosynthetic, holozoic, mixotrophic, saprotrophic
B4.2.7	Diverse nutrition in archaea
B4.2.8	Teeth and hominid diets
B4.2.9–10	Herbivores, plant defence, predators, prey
B4.2.11	Reaching light in forests
B4.2.12–13	Fundamental vs realized; competitive exclusion

Check that you can...

1. **Define** niche
2. **Classify** by oxygen need
3. **Compare** modes of nutrition
4. **Explain** herbivore, plant, predator and prey adaptations
5. **Distinguish** fundamental and realized niches
6. **Explain** competitive exclusion

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State the difference between an obligate anaerobe and a facultative anaerobe.
2. Describe saprotrophic nutrition.
3. Distinguish between a fundamental and a realized niche.

Image credits

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Euglena sp.jpg – Deuterostome – CC BY-SA 3.0

Mycena haematopus. (42029182981).jpg – Bernard Spragg. NZ from Christchurch, New Zealand – Public domain

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Human skull, with details showing the teeth. Etching. Wellcome V0008810.jpg – Author not stated – CC BY 4.0

Monarch caterpillar eating a milkweed leaf in Chippewa County, Michigan.jpg – USDAgov – Public domain

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Tillandsia sp. epiphytes on mahogany tree (Sanibel Island, Florida, USA).jpg – James St. John – CC BY 2.0

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